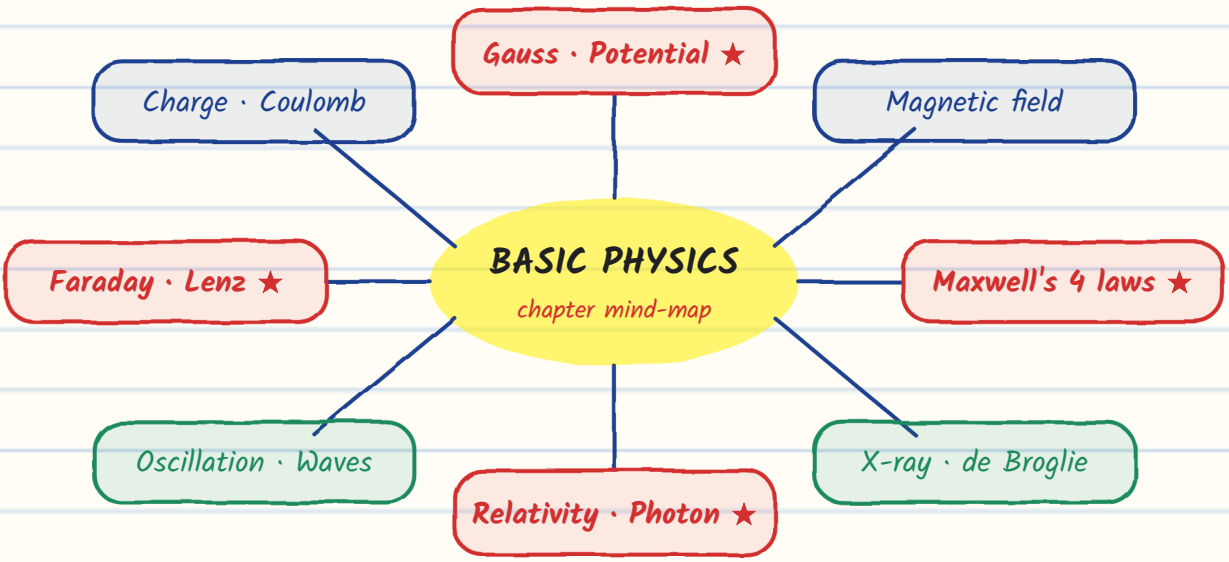
Basic Physics

Electromagnetism · Waves · Relativity · Quantum: exam-ready in 14 sheets



↑ the whole syllabus on one page: red = high-yield, green = formula-driven

➜ Syllabus coverage

Charge, Coulomb's law, electric field, Gauss's law, potential, magnetic field; Faraday's law, Maxwell's equations / waves & oscillations / special relativity / EM waves, photoelectric effect, quantum theory of light, X-ray & diffraction, Compton effect / de Broglie waves, phase & group velocity, wave function & wave equation.

High-yield list

① k, ε_0, μ_0, e, h values
② Gauss: $\Phi = q/\varepsilon_0$
③ Maxwell's 4 laws
④ γ, time dilation
⑤ $hf = \varphi + K_{max}$
⑥ $\lambda = h/p$, Compton shift

Legend yellow = key term · red circle = must-memorise value · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

➜ Contents

✓ Charge, Coulomb, electric field ..... p. 2
✓ Gauss's law, potential ..... p. 3
✓ Magnetic field, Faraday, Lenz ..... p. 4
✓ Maxwell's equations, EM spectrum ... p. 5
✓ Oscillations & SHM ..... p. 6
✓ Waves & wave equation ..... p. 7
✓ Special relativity ..... p. 8
✓ Photon, photoelectric effect ..... p. 9
✓ X-rays, Bragg ..... p. 10
✓ Compton, de Broglie ..... p. 11
✓ v_p, v_g, Ψ, Schrödinger ..... p. 12
✓ Formula & constants sheet ... p. 13
✓ Practice MCQs + written ..... p. 14

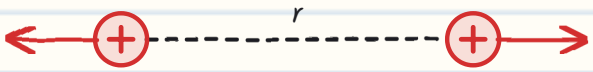
1 Electric Charge & Coulomb's Law

MCQ + Written

Definition **Charge (q):** the property of matter that makes it feel an electric force. SI unit: **coulomb (C)** = 1 A·s. Two kinds: + and −.

- **Quantised:** $q = \pm n e$, where $e = 1.602 \times 10^{-19}$ C
- **Conserved:** total charge of an isolated system never changes
- **Invariant:** same in every frame (unlike relativistic mass)
- 1 C ≈ 6.25×10^{18} electrons

↙ Like charges repel, unlike attract. Check the sign first!



like charges push apart

Coulomb's law

$F = k\, q_1 q_2 / r^2$

$k = 1/4\pi\varepsilon_0 = 8.99 \times 10^9\ N\cdot m^2/C^2$

$\varepsilon_0 = 8.854 \times 10^{-12}\ C^2/N\cdot m^2$

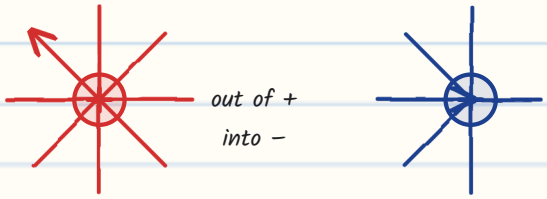
In a medium: $F_{medium} = F_{vacuum} / \varepsilon_r$ → water ($\varepsilon_r \approx 80$) cuts the force 80 times, which is why salt dissolves.

Exam trap Double r → F becomes ¼ (inverse square). Double *both* charges → 4F. Halve r → 4F.

Worked example ✍️ Two 1 µC charges, 30 cm apart: $F = 9\times10^9 \times (10^{-6})^2 / (0.3)^2 = 9\times10^{-3} / 0.09 =$ **0.1 N** (repulsive)

2 Electric Field (E)

Definition Force per unit positive test charge: $\mathbf{E = F / q_0}$. A vector; unit N/C = V/m. Point charge: $\mathbf{E = k\, q / r^2}$.

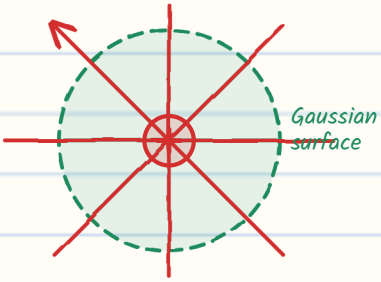


- Lines start on +, end on −
- They **never cross** (E has one direction per point)
- Closer lines = stronger field
- Meet a conductor **at 90°**; $E = 0$ inside it

Memory trick Field lines behave like water from a fountain: they flow **out of +** and drain **into −**. Analogy: gravity field $g = F/m$ ↔ electric field $E = F/q$.

3 Gauss's Law high-yield

Statement Total electric flux through any closed surface = enclosed charge / ε_0: $\oint E \cdot dA = q_{enc} / \varepsilon_0$



Flux $\Phi_E = EA \cos\theta$ ($N \cdot m^2/C$) = field lines piercing a surface.

↙ Flux does NOT depend on the size or shape of the surface, only on q inside!

Charge **outside** the surface → net flux **0** (every line in also goes out).

Charge distribution	Field E	Remember
Point charge / outside a sphere	kq/r^2	acts like a point charge
Infinite line (λ C/m)	$\lambda / 2\pi\varepsilon_0 r$	$\propto 1/r$
Infinite plane sheet (σ C/m²)	$\sigma / 2\varepsilon_0$	independent of distance
Just outside a conductor	σ / ε_0	twice the sheet value
Inside a conductor	**0**	charge sits on the surface

Classic MCQ Charge q at the centre of a cube → flux through **one face** = $q / 6\varepsilon_0$ (total q/ε_0 shared equally by 6 faces).

4 Electric Potential (V)

Definition Work done per unit positive charge to bring it from infinity: **V = W / q**. A scalar; unit volt = J/C. Point charge: $V = kq / r$ ($\propto 1/r$, not $1/r^2$)

- **E = −dV/dr**: the field points from high to low potential
- Energy: $\Delta U = q\Delta V$; **1 eV** = 1.602×10^{-19} J (energy of e through 1 V)
- **Equipotentials** ⊥ field lines: no work along them; a conductor is one equipotential

	Field E	Potential V
type	vector	scalar
point q	kq/r^2	kq/r
unit	N/C	J/C = V

Memory trick **E is the downhill slope of V**: a + charge rolls from high V to low V along E.

Worked example ✍ V at 10 cm from a 1 nC charge = $9\times10^9 \times 10^{-9} / 0.1$ = 90 V

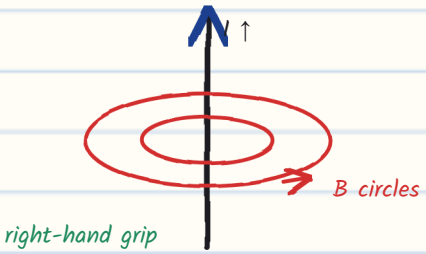
5 Magnetic Field (B)

Definition Region where a moving charge or current feels a force. Unit **tesla (T)** = N/(A·m); 1 T = 10^4 gauss · $\mu_0 = 4\pi \times 10^{-7}$ T·m/A

Lorentz force: $F = q(v \times B)$

$|F| = qvB \sin\theta$ · on a wire: $F = BIL \sin\theta$

$F \perp v \rightarrow$ **no work** is done; speed stays fixed, the path is a circle with $r = mv / qB$.



Source	B	Note
Long straight wire	$\mu_0 I / 2\pi r$	$\propto 1/r$
Centre of a circular loop	$\mu_0 I / 2R$	N turns → ×N
Long solenoid (n turns/m)	$\mu_0 n I$	uniform inside

Memory trick Fleming's **left** hand = **motor**: **F**irst finger = **F**ield, se**C**ond = **C**urrent, thu**M**b = **M**otion. Right hand = generator.

Exam trap Charge moving **parallel** to B ($\theta = 0°$) or at rest → magnetic force = **0**.

6 Faraday's Law & Lenz's Law high-yield

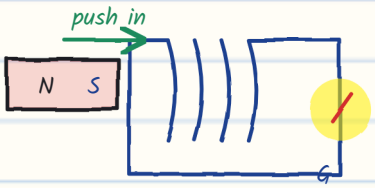
Faraday A changing magnetic flux induces an emf: $\varepsilon = -N \, d\Phi_B/dt$, where $\Phi_B = BA\cos\theta$ (weber, Wb = T·m²)

Lenz's law (the minus sign): the induced current opposes the change that causes it. This is energy conservation.

Change B, area A or angle θ → emf. Rotating coil = **AC generator**: $\varepsilon = NBA\omega \sin\omega t$

Moving rod: $\varepsilon = BLv$ (motional emf)

Uses: generator, **transformer** ($V_s/V_p = N_s/N_p$, **AC only**), induction cooker, card readers, eddy-current brakes.



needle kicks only while moving

Worked example 100 turns, flux 0.02 Wb → 0 in 0.1 s: $|\varepsilon| = 100 \times 0.02 / 0.1 =$ **20 V**

7 Maxwell's Equations (1865) high-yield

Four equations that **unify electricity, magnetism and light**:

#	Law	Integral form	Meaning
1	Gauss (E)	$\oint E\cdot dA = q / \varepsilon_0$	charges create E
2	Gauss (B)	$\oint B\cdot dA = 0$	**no magnetic monopoles**
3	Faraday	$\oint E\cdot dl = -d\Phi_B/dt$	changing B → E
4	Ampère–Maxwell	$\oint B\cdot dl = \mu_0 I + \mu_0\varepsilon_0\, d\Phi_E/dt$	current & changing E → B

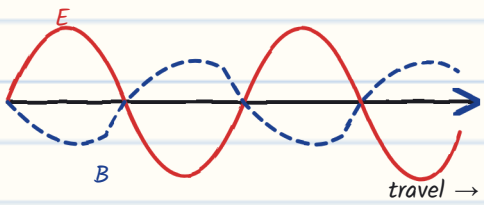
Differential: $\nabla\cdot E = \rho/\varepsilon_0$ · $\nabla\cdot B = 0$ · $\nabla\times E = -\partial B/\partial t$ · $\nabla\times B = \mu_0 J + \mu_0\varepsilon_0\,\partial E/\partial t$

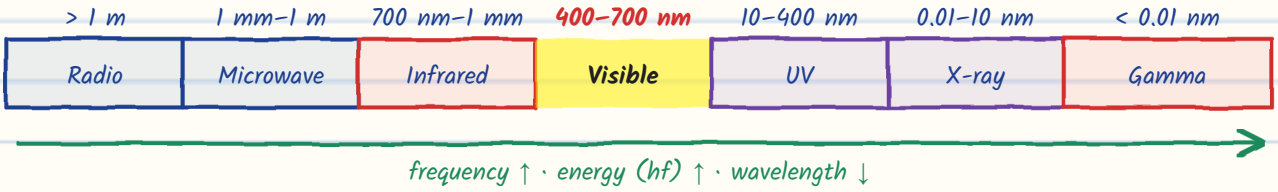
Displacement current $I_d = \varepsilon_0\, d\Phi_E/dt$: Maxwell's own addition. It lets "current" flow across a **charging capacitor's gap**, and it is what makes EM waves possible.

Memory trick Order **G-G-F-A**: two **divergences** (sources: charge yes, monopole no) and two **curls** (circulation: Faraday, Ampère).

8 EM Waves & the Spectrum

- Maxwell predicted them; **Hertz (1887–88)** produced them
- Speed $c = 1/\sqrt{(\mu_0\varepsilon_0)} = 3 \times 10^8$ m/s
- **E ⊥ B ⊥ direction**: **transverse**, $E/B = c$
- Need **no medium**; carry energy & momentum





- **Microwave**: Wi-Fi (2.4/5 GHz), mobile, radar, oven
- **Infrared**: TV remote, thermal camera; optical fibre uses near-IR (1310/1550 nm)
- **UV**: sterilisation, forged-note check
- **X-ray**: medical imaging, crystal structure · **Gamma**: radiotherapy

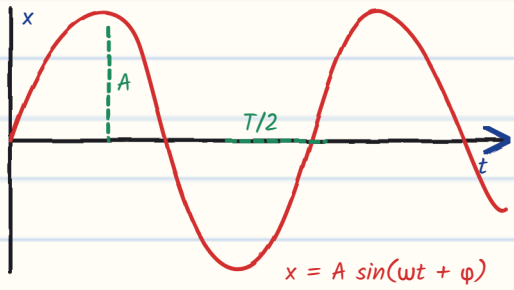
Memory trick "**R**aging **M**artians **I**nvaded **V**enus **U**sing **X**-ray **G**uns" = increasing frequency.

Exam trap In vacuum **every** EM wave travels at the same c; they differ only in f and λ. Sound is **not** an EM wave.

9 Oscillations & Simple Harmonic Motion

SHM Periodic motion in which the restoring force ∝ displacement and points back to the mean position: $F = -kx \Rightarrow a = -\omega^2 x$

Term	Meaning
Amplitude A	max displacement
Period T	time of 1 oscillation
Frequency f	$f = 1/T$ (Hz)
Angular freq. ω	$\omega = 2\pi f = 2\pi/T$
Phase φ	starting position



Key equations

$x = A\sin(\omega t + \varphi)$ $v = \omega\sqrt{(A^2 - x^2)}$

$v_{max} = A\omega$ (at mean position) $a_{max} = A\omega^2$ (at the extremes)

Total energy $E = \frac{1}{2}kA^2$ (constant) KE max at mean, PE max at extremes

Oscillator	Period	Depends on
Spring–mass	$T = 2\pi\sqrt{(m/k)}$	m, k (not g)
Simple pendulum (small θ)	$T = 2\pi\sqrt{(L/g)}$	L, g (not mass, not amplitude)
LC circuit (electrical)	$f = 1 / 2\pi\sqrt{(LC)}$	L, C → radio tuning

Worked example ✍ $k = 100$ N/m, $m = 1$ kg → $\omega = \sqrt{(k/m)} = 10$ rad/s, $T = 2\pi/10 \approx$ **0.63 s**

Exam trap Pendulum on the Moon ($g/6$) → T becomes **$\sqrt{6} \approx 2.45$×**. Double the bob's mass → **no change**. Length ×4 → T ×2.

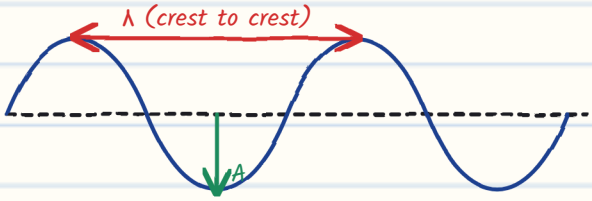
➜ Damped, forced & resonance

- **Damped**: friction drains energy → amplitude decays (shock absorbers)
- **Forced**: an external periodic force drives the system
- **Resonance**: driving frequency = natural frequency → **maximum amplitude** (radio tuning, pushing a swing in step)

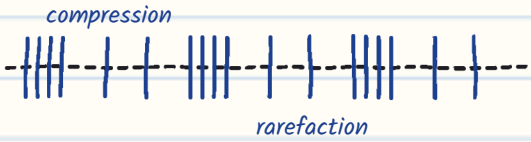
Memory trick **Mean = max speed, Extreme = max acceleration.** Like a swing: fastest at the bottom, stops and turns at the ends.

10 Waves & the Wave Equation

Definition A wave is a travelling disturbance that carries energy, not matter.



TRANSVERSE: particles ⊥ travel



LONGITUDINAL: particles ∥ travel

Type	Examples	Key fact
Mechanical	sound, water, string	need a medium
Electromagnetic	light, radio, X-ray	travel in vacuum
Transverse	string, all EM waves	can be **polarised**
Longitudinal	sound in air	cannot be polarised

Wave formulas

$v = f\lambda$ $k = 2\pi/\lambda$ (wave number), $v = \omega/k$

$y(x, t) = A\sin(kx - \omega t)$ String: $v = \sqrt{(T/\mu)}$

Wave equation: $\partial^2y/\partial x^2 = (1/v^2)\,\partial^2y/\partial t^2$; any shape $f(x - vt)$ solves it.

- Sound in air ≈ 343 m/s (20 °C; 332 at 0 °C)
- $v_{solid} > v_{liquid} > v_{gas}$; no sound in vacuum
- Constructive: path difference $n\lambda$; destructive: $(n + ½)\lambda$
- Standing wave: node to node = **λ/2**
- Beats: $f_{beat} = |f_1 - f_2|$
- Doppler: source approaching → higher pitch

Audible range 20 Hz – 20 kHz, below = **infrasonic** (earthquakes), above = **ultrasonic** (SONAR, medical scans). Intensity ∝ A^2: double the amplitude → 4× the energy.

Exam trap When a wave enters a new medium its **frequency stays the same**; only speed and λ change.

IT link Wi-Fi at 2.4 GHz: $\lambda = c/f = 3\times10^8 / 2.4\times10^9 =$ 12.5 cm. FM 100 MHz: λ = **3 m** → that is why antennas are sized in λ/2 or λ/4.

11 Special Theory of Relativity

Einstein 1905

Postulates ① The laws of physics are the same in every inertial frame. ② The speed of light in vacuum, $c = 3 \times 10^8$ m/s, is the same for all observers, whatever the source's motion.

Everything follows from the **Lorentz factor**: $\gamma = 1/\sqrt{(1 - v^2/c^2)}$ (always ≥ 1)

↙ Michelson–Morley (1887): no "ether" found, supporting postulate 2

Effect	Formula	Meaning
Time dilation	$\Delta t = \gamma\, \Delta t_0$	moving clock runs slow
Length contraction	$L = L_0 / \gamma$	only along the motion
Relativistic mass	$m = \gamma\, m_0$	$\to \infty$ as $v \to c$
Mass–energy	$E = mc^2$	rest energy m_0c^2
Energy–momentum	$E^2 = (pc)^2 + (m_0c^2)^2$	photon: $E = pc$

v	0.6c	0.8c	0.866c	0.99c
γ	1.25	1.67	2	7.09



Muon proof ✍ Lifetime at rest 2.2 μs. At 0.99c: $\Delta t = 7.09 \times 2.2 \approx$ 15.6 μs, so muons from the upper atmosphere reach the ground.

$E = mc^2$ ✍ 1 g of mass → $E = 10^{-3} \times (3\times10^8)^2 = 9 \times 10^{13}$ J; the basis of nuclear power.

Exam trap Nothing with mass reaches c: $u = (u' + v)/(1 + u'v/c^2) \Rightarrow c +$ anything $= c$. t_0, L_0 = **proper** values (rest frame). Charge never changes with speed.

Memory trick "**Moving clocks run slow, moving rulers shrink**": γ multiplies time, divides length.

IT link GPS satellite clocks need relativity corrections (≈ 38 μs/day net), or positions drift by km per day.

12 Quantum Theory of Light

Planck 1900 · Einstein 1905

Photon Light travels as packets (quanta) of energy $E = hf = hc/\lambda$, momentum $p = h/\lambda = E/c$, rest mass **zero**, speed c.

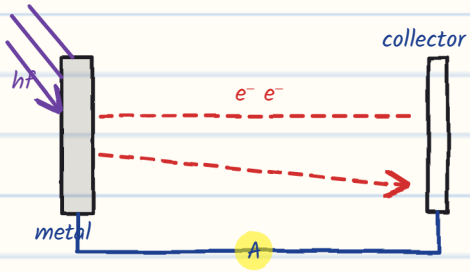
$h = 6.626 \times 10^{-34}$ J·s · shortcut $hc = 1240$ eV·nm → $E(eV) = 1240/\lambda(nm)$: red 620 nm = **2 eV**

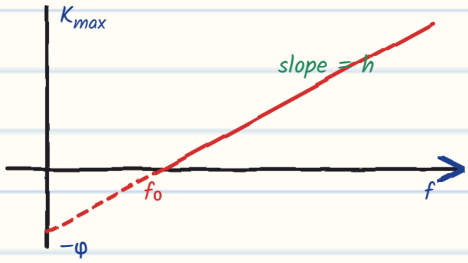
Wave–particle duality: interference & diffraction → wave; photoelectric effect & Compton effect → particle.

13 Photoelectric Effect

high-yield

Definition Emission of electrons from a metal surface when light of high enough frequency falls on it. Observed by Hertz (1887); explained by Einstein (1905, Nobel 1921).





Einstein's equation $hf = \varphi + K_{max}$ · $K_{max} = eV_0$ (V_0 = stopping potential) · threshold $f_0 = \varphi/h$, $\lambda_0 = hc/\varphi$

You increase…	Photocurrent	K_{max} / V_0
Intensity (f fixed, $f > f_0$)	**increases** (more photons)	no change
Frequency (intensity fixed)	≈ no change	**increases** linearly
Anything, but $f < f_0$	no emission, however bright · emission is instant	

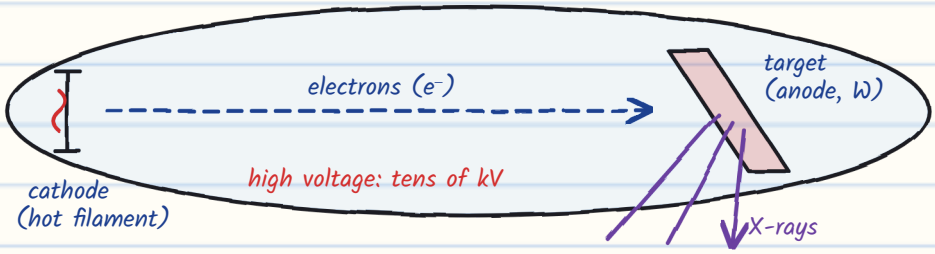
Exam trap Brighter light means **more** electrons, not **faster** ones. Classical wave theory fails on all three points: threshold, K_{max} vs f, no time lag.

Worked example ✍️ 400 nm light on sodium ($\varphi = 2.3$ eV): $E = 1240/400 = 3.1$ eV → $K_{max} = 3.1 - 2.3 =$ **0.8 eV** → $V_0 = 0.8$ V

IT link: photodiodes in optical-fibre receivers, CCD/CMOS camera sensors and solar cells all work on photon absorption.

14 X-Rays (Röntgen 1895)

Definition EM waves of $\lambda \approx 0.01 - 10$ nm, produced when fast electrons strike a heavy-metal target (usually **tungsten**).



→ Two parts of the spectrum

- **Continuous** (bremsstrahlung = "braking radiation"): electrons slowing down near nuclei
- **Characteristic lines** (K_α, K_β): inner-shell electron jumps → fingerprint of the target element

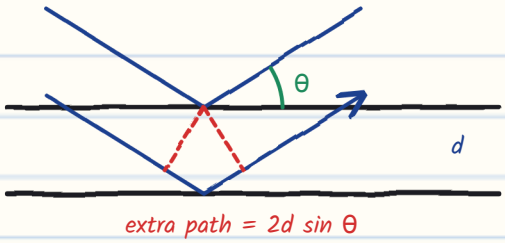
→ Properties

- Penetrating & ionising; blacken photographic film
- **Not deflected** by E or B fields (uncharged)
- Hard X-rays = short λ = more penetrating

Cut-off wavelength All of eV goes into one photon: $\lambda_{min} = hc / eV$ → $\lambda_{min}(nm) = 1240 / V(volts)$. 30 kV tube → $\lambda_{min} \approx$ **0.041 nm**

15 X-Ray Diffraction & Bragg's Law

Atomic spacing in crystals (~0.1 nm) ≈ X-ray λ, so a crystal acts as a 3-D diffraction grating (Laue 1912; W.H. & W.L. Bragg 1913).



Bragg's law

$2d \sin\theta = n\lambda$ ($n = 1, 2, 3 \ldots$)

Uses: finding crystal structure; Rosalind Franklin's X-ray image (1952) revealed the **DNA double helix**.

Exam trap In Bragg's law θ is the **glancing angle** (ray to the crystal plane), **not** the angle to the normal. The largest possible λ is **2d** ($\sin\theta \leq 1$).

Worked example ✍ $d = 0.28$ nm, first order at $\theta = 15°$: $\lambda = 2 \times 0.28 \times \sin 15° = 0.56 \times 0.259 \approx$ **0.145 nm**

16 Compton Effect

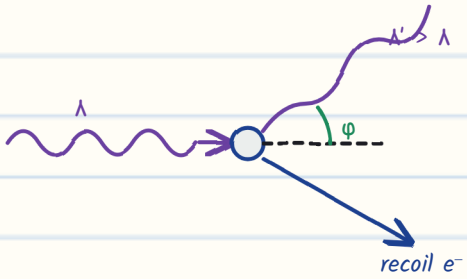
A.H. Compton 1923

Definition An X-ray photon scattered by a nearly free electron comes out with a longer wavelength. The photon behaves like a billiard ball with momentum $p = h/\lambda$.

Compton shift

$\Delta\lambda = \lambda' - \lambda = (h / m_e c)(1 - \cos\varphi)$

$h / m_e c = 2.43$ pm (Compton wavelength)



φ	0°	90°	180°
Δλ	0	2.43 pm	4.85 pm (max)

- Shift depends **only on φ**, not on λ or the target material
- It proves photons carry momentum: strong evidence for the particle nature of light

Exam trap Why X-rays, not visible light? Δλ is only a few pm, which is negligible next to 500 nm but large next to 0.07 nm.

17 de Broglie Waves (Matter Waves)

1924

Hypothesis If light (a wave) acts like particles, then particles act like waves: $\lambda = h / p = h / mv = h / \sqrt{(2mK)}$

Electron through V volts

$\lambda = 1.226 / \sqrt{V}$ nm

100 V → λ ≈ **0.123 nm**, the same scale as X-rays → electrons diffract from crystals!

Why don't we see it? Cricket ball, 0.1 kg at 10 m/s: $\lambda = 6.6\times10^{-34} / 1 = 6.6 \times 10^{-34}$ m, far too small to detect.

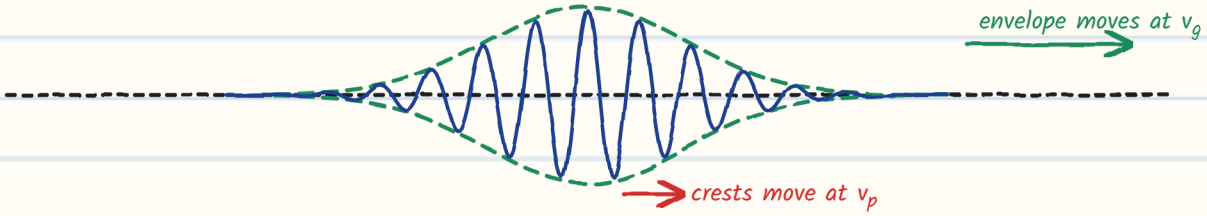
- **Davisson–Germer (1927):** electron diffraction from a nickel crystal (54 V electrons) confirmed matter waves
- **Bohr orbits:** $2\pi r = n\lambda$ (whole wavelengths fit) → quantised orbits

Memory trick **Light → particle** (photoelectric, Compton); **particle → wave** (de Broglie, Davisson–Germer). Same h links both.

IT link: electron microscopes use tiny electron λ to image chip features far smaller than visible light can resolve.

18 Phase Velocity & Group Velocity

A localised particle = a **wave packet**, i.e. many waves of slightly different k added together.



	Phase velocity v_p	Group velocity v_g
definition	$\omega / k = E / p$	$d\omega / dk = dE / dp$
matter wave	$c^2 / v > c$ (!)	= v (particle speed)
carries	no information	energy & information

Product: $v_p \cdot v_g = c^2$

Light in vacuum: $v_p = v_g = c$

Heisenberg: $\Delta x \cdot \Delta p \geq \hbar/2$, $\hbar = h/2\pi$

Exam trap $v_p > c$ does **not** break relativity, because no signal travels at v_p. Group velocity = dE/dp (some texts misprint dp/dm).

19 Wave Function & Schrödinger Equation 1926

Wave function Ψ(x, t) Holds everything knowable about a particle. Ψ itself is not measurable and may be complex, but $|\Psi|^2$ = probability density (Born rule).

- ✓ Normalised: $\int |\Psi|^2 \, dx = 1$
- ✓ Single-valued, finite, continuous
- ✓ $d\Psi/dx$ continuous (where V is finite)
- ✓ Measurement "collapses" Ψ

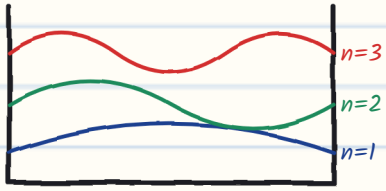
Schrödinger equation

Time-dependent: $i\hbar \, \partial\Psi/\partial t = -(\hbar^2/2m) \, \partial^2\Psi/\partial x^2 + V\Psi$

Time-independent (steady states): $-(\hbar^2/2m) \, d^2\psi/dx^2 + V\psi = E\psi$

Particle in a 1-D box (width L): $E_n = n^2h^2 / 8mL^2$

$n = 1, 2, 3 \ldots$



Energy is **quantised**; n = 1 is not zero (zero-point energy). **IT link:** quantum **tunnelling** programs and erases NAND flash cells (SSDs, pen drives).

Night-Before-the-Exam Sheet

Constants: memorise these

$c = 3 \times 10^8$ m/s
$e = 1.602 \times 10^{-19}$ C
$m_e = 9.11 \times 10^{-31}$ kg
$m_p = 1.673 \times 10^{-27}$ kg
$h = 6.626 \times 10^{-34}$ J·s
$\hbar = 1.055 \times 10^{-34}$ J·s
$k = 8.99 \times 10^9$ N·m²/C²
$\varepsilon_0 = 8.854 \times 10^{-12}$ C²/N·m²
$\mu_0 = 4\pi \times 10^{-7}$ T·m/A
1 eV = 1.602×10^{-19} J
$hc = 1240$ eV·nm
$h/m_e c = 2.43$ pm

Formula sheet

$F = kq_1q_2/r^2$ · $E = F/q$
$V = kq/r$ · $E = -dV/dr$
$\oint E \cdot dA = q/\varepsilon_0$ · sheet $\sigma/2\varepsilon_0$
$F = qvB \sin\theta$ · $F = BIL \sin\theta$
$B = \mu_0 I/2\pi r$ · solenoid $\mu_0 nI$
$\varepsilon = -N\, d\Phi/dt$ · $\varepsilon = BLv$
$c = 1/\sqrt{(\mu_0\varepsilon_0)}$ · $v = f\lambda$
$T = 2\pi\sqrt{(m/k)}$ · $2\pi\sqrt{(L/g)}$
$\gamma = 1/\sqrt{(1 - v^2/c^2)}$
$\Delta t = \gamma \Delta t_0$ · $L = L_0/\gamma$ · $E = mc^2$
$E = hf$ · $hf = \varphi + K_{max}$
$\lambda_{min} = hc/eV$ · $2d \sin\theta = n\lambda$
$\Delta\lambda = (h/m_e c)(1 - \cos\varphi)$
$\lambda = h/p$ · e^-: $1.226/\sqrt{V}$ nm
$v_p = \omega/k$ · $v_g = d\omega/dk$
$E_n = n^2h^2/8mL^2$

→ Top 12 one-liners for MCQs

1. Electric flux unit: N·m²/C
2. Inside a conductor: E = 0, V constant
3. $\oint B \cdot dA = 0 \rightarrow$ no monopoles
4. Lenz's law = energy conservation
5. Displacement current: Maxwell
6. Pendulum T is independent of mass
7. Sound (longitudinal) cannot be polarised
8. $\gamma(0.6c) = 1.25$, $\gamma(0.8c) = 1.67$
9. K_{max} depends on f, not intensity
10. Compton shift max at 180° = 4.85 pm
11. Davisson–Germer proved matter waves
12. $|\Psi|^2$ = probability density

Top traps $1/r^2$ (E, F) vs $1/r$ (V) · θ in Bragg = glancing angle · brighter ≠ faster electrons · $v_p > c$ is allowed · frequency is unchanged when entering a new medium

All tricks together Fountain (field lines) · E = downhill slope of V · **F**ield-**C**urrent-**M**otion (left hand = motor) · **GGFA** (Maxwell) · **RMIVUXG** (spectrum) · mean = max speed · moving clocks slow, rulers shrink

✓ Practice MCQs (exam pattern, 10 marks)

try first, then check

1. If the distance between two point charges is tripled, the force becomes:
(a) F/9 (b) F/3
(c) 3F (d) 9F

2. SI unit of magnetic flux:
(a) tesla (b) henry
(c) weber (d) gauss

3. Which Maxwell equation rules out magnetic monopoles?
(a) $\oint E \cdot dA = q/\varepsilon_0$ (b) $\oint E \cdot dl = -d\Phi_B/dt$
(c) $\oint B \cdot dl = \mu_0 I$ (d) $\oint B \cdot dA = 0$

4. Wavelength of a 3 GHz microwave:
(a) 1 cm (b) 10 cm
(c) 1 m (d) 10 m

5. A pendulum's length is made 4 times. Its period becomes:
(a) 2T (b) T/2
(c) 4T (d) T

6. Lorentz factor γ at v = 0.8c:
(a) 1.25 (b) 2
(c) 1.67 (d) 0.6

7. Photoelectric effect, $f > f_0$ fixed. Raising the intensity increases:
(a) K_{max} (b) photocurrent
(c) V_0 (d) f_0

8. Energy of a 620 nm photon:
(a) 1 eV (b) 3 eV
(c) 6.2 eV (d) 2 eV

9. Compton shift is maximum at a scattering angle of:
(a) 0° (b) 45°
(c) 180° (d) 90°

10. de Broglie λ of an electron accelerated through 100 V:
(a) 0.123 nm (b) 0.012 nm
(c) 1.23 nm (d) 12.3 nm

Answer key → 1-a · 2-c · 3-d · 4-b · 5-a · 6-c · 7-b · 8-d · 9-c · 10-a

Written question State Einstein's photoelectric equation. Light of wavelength 300 nm falls on a metal of work function 2.5 eV. Find (i) K_{max}, (ii) the stopping potential, (iii) the threshold wavelength. 8 marks

1. **Equation:** $hf = \varphi + K_{max}$: photon energy = work function + max kinetic energy of the ejected electron. 2
2. **Photon energy:** $E = hc/\lambda = 1240 / 300 =$ **4.13 eV** 2
3. **(i)** $K_{max} = 4.13 - 2.5 =$ **1.63 eV** $(= 2.6 \times 10^{-19}$ J) 2
4. **(ii)** $eV_0 = K_{max} \rightarrow V_0 =$ **1.63 V** · **(iii)** $\lambda_0 = hc/\varphi = 1240 / 2.5 =$ **496 nm** 2

→ Before the exam ✓

- ✓ Can write all 4 Maxwell equations + meaning
- ✓ Can draw E-field lines & Gauss surfaces
- ✓ Know γ table & both relativity effects
- ✓ Can solve $hf = \varphi + K_{max}$ in eV units
- ✓ Know Bragg, Compton, de Broglie formulas

Also practise (written):
① Derive $E = kq/r^2$ from Gauss's law
② Explain displacement current
③ Time dilation with the muon example
④ Compton shift derivation (outline)